

The schematic diagram illustrates the power input section of a PCB, showing the flow from J1 (BAT1_9V) through various components like diodes (D1, D5), capacitors (C1, C2, C3), and a voltage regulator (U3) to the VIN_OR and VM pins.

POWER INPUT / OR-ING / FUSE / LDO

The diagram shows the following components and connections:

- J1 (BAT1_9V):** Input connector with pins 1 (VIN) and 2 (GND).
- D1:** Diode (SS34) connected between VBAT1_IN and VIN_OR.
- J6 (BAT2_9V):** Input connector with pins 1 (VIN) and 2 (GND).
- D5:** Diode (SS34) connected between VBAT2_IN and VIN_OR.
- J5 (DC_9-12V):** Input connector with pins 1 (VIN_DC), 2 (GND), and 3 (GND).
- D4:** Diode (SS34) connected between VIN_DC and VIN_OR.
- F1:** Fuse (PPTC, 1A MF-R110) connected between VIN_OR and the input of the voltage regulator U3.
- TVS1:** TVS diode (SMAJ15A) connected between VIN_DC and GND.
- TVS2:** TVS diode (SMAJ15A) connected between VM and GND.
- C1:** Capacitor (1000uF 25V) connected between VIN and GND.
- U3:** Voltage regulator (LDO, 1.05C1570BPS302E/CB) connected between VIN and VM.
- C2:** Capacitor (1uF X7R 25V) connected between VM and GND.
- C3:** Capacitor (1uF X7R 16V) connected between VM and GND.

BUTTONS (RC DEBOUNCE + ESD SERIES R)

The diagram illustrates the internal wiring and protection for two buttons, SW1 and SW2, connected to an MCU.

SW1 (UP Button):

- Internal wiring: A 4-pin component with pins 1 and 2 connected to GND, and pins 3 and 4 connected to GND.
- MCU connection: The button is connected to the MCU via a 10k pull-up resistor (R11) to +5V and a 00R resistor (R9) to GND.
- ESD protection: A 10k resistor (R10) is connected between the button and GND, and a 100nF capacitor (C10) is connected between the button and GND.

SW2 (SET Button):

- Internal wiring: A 4-pin component with pins 1 and 2 connected to GND, and pins 3 and 4 connected to GND.
- MCU connection: The button is connected to the MCU via a 10k pull-up resistor (R12) to +5V and a 00R resistor (R10) to GND.
- ESD protection: A 10k resistor (R11) is connected between the button and GND, and a 100nF capacitor (C11) is connected between the button and GND.

Netlist authority: NETLIST.m Rev 2026-08-11
 All 9V/12V sources: Or-ed (5534) then fused (F1): C1+C19 are 2x1000uF VM reservoir/s
 HW fail-safe: TPS3700 VM supervisor -> /VM_OK selects two SN74LVC1G3157 muxes for
 Attiny84A + DS5231M(+supercap) + TM1637 + DRV8871 bistable solenoid
GreenGuard USA
 Sheet: /
 File: co2_timer_v5.kicad_sch

Title: GreenGuard Co2 Trap Timer		
Size: A2	Date: 2026-08-11	Rev: 5.0
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